

An FPGA-Based High-Frequency Trading System for 10 Gigabit Ethernet with a Latency of 433 ns

Yi-Chieh Kao

Department of Electrical Engineering
National Tsing Hua University
Hsinchu, R.O.C.
jjhsb29@gmail.com

Hung-An Chen

Department of Electrical Engineering
National Tsing Hua University
Hsinchu, R.O.C.
a229295091@gmail.com

Hsi-Pin Ma

Department of Electrical Engineering
National Tsing Hua University
Hsinchu, R.O.C.
hp@ee.nthu.edu.tw

Abstract—High-frequency trading (HFT) systems require extremely low latency in response to market feeds to make profits. In order to reduce latency, the system is implemented with a 10 gigabit Ethernet physical transceiver with a low latency of 25 ns, custom network stack parsing and packaging, partial financial protocol decoding and encoding, order book handling and custom trading strategy. The hardware test and functional verification of the trading system use Taiwan futures trading environment provided by Yuanta Futures. The latency from the internal market packet analysis to the ordering packet triggered is approximately 433 ns.

Keywords—high-frequency trading, hardware acceleration, futures

I. INTRODUCTION

A. Background

During the last two decades, technological and regulatory escalation have revolutionized the way that financial markets work. Markets have changed from exclusively human-driven systems to mainly computerized trading.

High-frequency trading (HFT) refers to the use of extremely low-delay and sophisticated computer programs to carry out a huge amount of trading executions in a quite short time and accumulate profits through extremely frequent transactions and small spreads. Therefore, numerous orders are sent to the financial markets with round-trip time estimated within microseconds [1]. However, the few traders whose request first be sent to exchange can only profit from a given opportunity. That's the main reason why latency matters.

Due to the huge profits carried out by HFT, more and more traders engage in the HFT space and attempt to seize these opportunities, which makes the trading speed competition more and more fierce.

B. Main Features

- **Network stack** decoder decodes the received frames in an efficient pipelined design on FPGA to reduce high latency of the common software-based system which the Ethernet frame is received by a network interface card (NIC) and be decoded by operating system.
- Custom financial protocol decoder for Taiwan futures market is implemented to simplify the original parsing process, focus on the messages that would affect the trading behavior only.
- Order book proposed by this paper can handle **top five price and size** and a derivative price for both bid and ask of a single product. It minimizes the latency of book handling compares with a universal order book. An expandable hardware architecture is presented that

can easily handle multiple product by adding more order book modules.

- Custom TCP offload engine is implemented to provide basic functionalities of TCP connection and disconnection.
- Functions of manual order placement and TCP operation based on different connection parameters through DMA subsystem.

II. HARDWARE ARCHITECTURE

The architecture of the proposed 10 gigabit Ethernet high-frequency trading system is shown in Fig. 1.

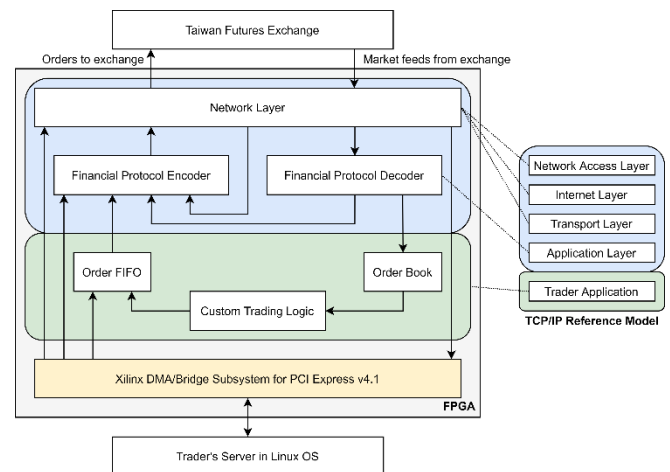


Fig. 1. The architecture of FPGA-based HFT system.

The first part is **network layer module**, including the functionality of network access layer, internet layer and transport layer in TCP/IP reference model. It receives or transmits packets from or to network and offloads the TCP/IP or UDP/IP packets to financial protocol decoder.

The second part is **financial protocol decoder and encoder**. The financial protocol decoder performs two types of financial protocol parsing: TAIFEX message protocol (TMP) [2] and Taiwan tick-by-tick data protocol [3]. It processes the payload transferred by the network layer into the financial market information, which upgrades the current status of specific commodities in the order book and sends into financial protocol encoder for triggering the corresponding connection process of login to the subsystem of exchange. The financial protocol encoder transforms the manual order request made by trader himself or automatic custom trading strategy to a TCP packet.

Afterwards, the order book is used to handle top five bid/ask price of specific commodities. The custom trading logic, which is a simple strategy for arbitrage in the system,

This research is supported in part by the Ministry of Science and Technology, Taiwan, R.O.C. under grant 109-2622-E-007-010-CC2 and in part by Yuanta Futures.

places orders according to the information in the order book to earn the spread. Finally, an order FIFO is added into the system for storing the order information from the DMA subsystem or the custom trading logic.

For the tasks that are not timing critical, a DMA subsystem is used to provide the functionality of data transfer between and host central processing unit (CPU). Therefore, these task can be finished by the host CPU, such as supervision and reporting. This allows the proposed system to avoid the unpredictable of PCIe.

III. IMPLEMENTATION RESULT

A. Functional Verification Method

1) *Bypass of UDP/TCP packets*: The block diagram of the test is shown in Fig. 2. I store the data which is output form the receiving side of Xilinx PCS/PMA IP core to register for timing closure. After that, the data buffered by register is sent to the transmitting end of the IP core and transmitted to the NIC of another host CPU to collect and record the packets sent by HFT system in order to judge whether the packets received by the FPGA-based HFT system are consistent with the NIC of trader server.

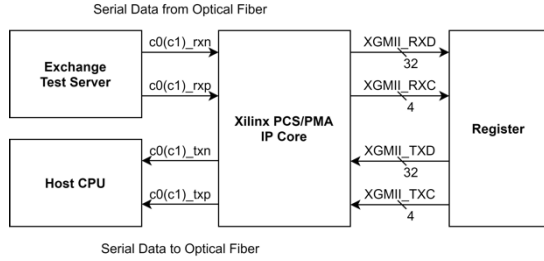


Fig. 2. Block diagram of bypass of UDP/TCP packets.

2) *Offloading of UDP/TCP packets*: The block diagram is shown in Fig. 3. The network layer module parses the serial data into payload and passes it to the financial protocol decoder for further processing. Afterwards, the financial protocol decoder stores the decoded message in the FIFO for Xilinx DMA subsystem to read through the C2H channel. Finally, host CPU fetches the data stored in the FIFO via PCIe.

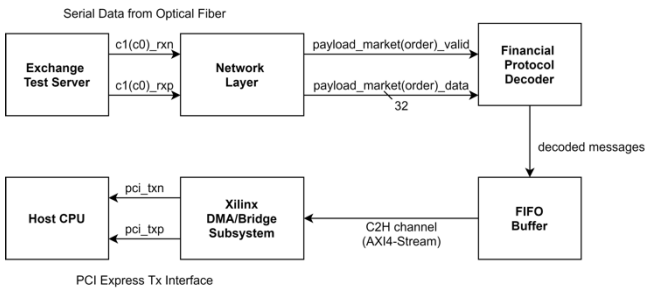


Fig. 3. Block diagram of UDP/TCP offloading.

3) *Connection function with exchange*: The block diagram is shown in Fig. 4. The network layer is responsible for the TCP three-way handshake while the TMP packets are processed by financial protocol encoder respectively. Then, two kinds of packets are stored in the FIFO and sent to the exchange by the network layer as a transmitter.

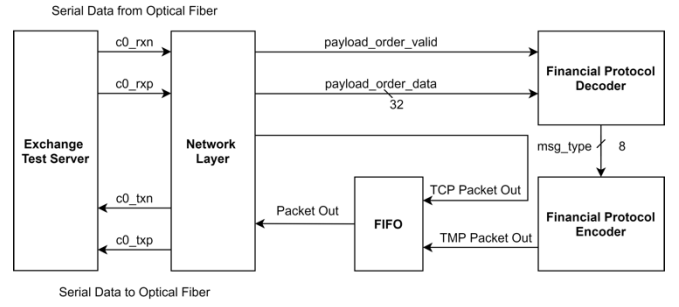


Fig. 4. Block diagram of connection verification.

4) *Function of manual order placement*: The block diagram is shown in Fig. 5. the trader issues corresponding instruction to provide information about the order he wants to place, including product ID, price, quantity and so on, through Xilinx DMA subsystem on the host CPU. The manual order information will be stored in the FIFO buffer until the financial protocol encoder is ready to process.

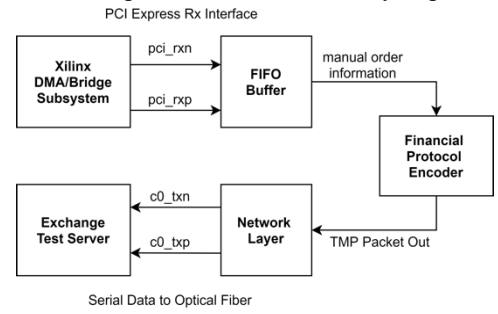


Fig. 5. Block diagram of verification for manual order placement.

B. Utilization of Hardware Resources

A Xilinx Kintex Ultrascale XCKU115-2-FLVA1517E FPGA board is used for the hardware implementation of 10 gigabit Ethernet high-frequency trading system. The utilization resource is shown in Table I.

TABLE I. RESOURCE UTILIZATION

Resource	Utilization	Available	Utilization (%)
LUT	70301	663360	10.60
LUTRAM	14745	293760	5.02
FF	73575	1326720	5.55
BRAM	66	2160	3.06
DSP	1	5520	0.02
IO	48	624	7.69
GT	10	48	20.83
BUFG	6	1248	0.48
MMCM	1	24	4.17
PCIe	1	6	16.67

C. Performance Evaluation

Fig. 6. shows the latency and the system flow of receiving an UDP data then transmit an order packet in the system. The latency of each functional block and the end-to-end latency is estimated according to the Vivado simulator, I developed a simulation environment that simulate the behavior of futures

market feed handler and trading server. The functional block with the **largest latency is network layer which contains the 10 gigabit 32-bit Ethernet PCS/PMA IP** core. The latency of 32-bit 10 gigabit PCS/PMA IP core is 25 ns on both transmit and receive path. Thus, the end-to-end latency in the FPGA-based HFT system from network layer Rx to Tx is approximately 433 ns.

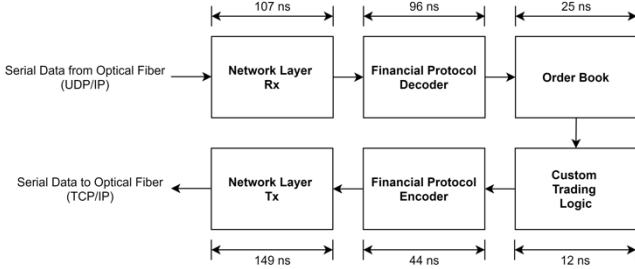


Fig. 6. Latency of each functional blocks in end-to-end path.

Table II compares the latency of different system with previous works.

TABLE II. LATENCY COMPARISON. [UNIT: μ S]

System Block	Tang [4]	Lockwood [1]	Boutros [5]	Proposed Work
Networking Rx	0.8	0.4	0.338	0.107
Financial Protocol Decoding	0.5	0.1	0.058	0.096
Order Book Handling	-	-	0.077	0.025
Trading Logic	-	-	-	0.012
Financial Protocol Encoding	-	0.1	0.058	0.044
Networking Tx	-	0.4	0.338	0.149

Because of the network layer that I proposed in this paper only implement UDP/IP, TCP/IP and ARP instead of a general purpose network stack in other prior works so the latency is lowest among all. For the financial protocol decoder, the standard protocol that is applied to other works is the financial information exchange (FIX) protocol adapted for streaming (FAST) which is used by multiple exchanges to distribute their market data [6]. Different from FAST, there isn't any encryption method in TMP and tick-by-tick data protocol. The order book I proposed is based on single product usage with 6 entries for bid/ask each can increase the size of order book by adding more order book modules. All order book modules are run in parallel.

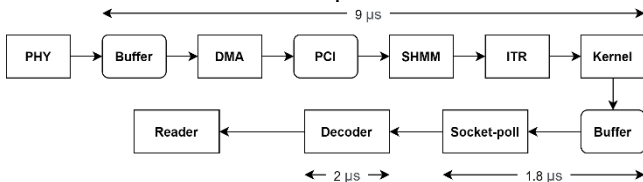


Fig. 7. Latency of a standard software-based HFT system [7].

In [7], a single frame reception of a typical software-based HFT system has been measured. Measured latencies for the NIC part is 9 μ s, for the kernel space 1.8 μ s and for user space 2 μ s. This results in an aggregated of 12.8 μ s. And the overall latencies of over 30 μ s of the frame reception and transmission.

Fig. 7 shows the flow of the messages in such a standard system.

Table 3.3 shows the HFT system latency before and after acceleration. The latency of software-based HFT system is measured in [7]. A software-based network layer struggle with the operating system's network stack and the unpredictable latency of PCI. As a result, the latency of network layer was decreased by over two orders of magnitude compares to the software-based network layer.

TABLE III. LATENCY BEFORE/AFTER THE ACCELERATION. [UNIT: μ S]

System Block	Software-based HFT System [7]	Proposed Work
Networking Rx	10.8	0.107
Financial Protocol Decoding	2	0.096
Order Book Handling	20	0.025
Trading Logic	X	0.012
Financial Protocol Encoding	2	0.044
Networking Tx	10.8	0.149
Overall	45.6+X	0.433

Because the latency of custom trading logic varies with different strategies, I use the variable X to represent it. A software-based uses external memory to store a large size of order book, while the order book that I proposed use on-chip memory to store a single product's top five bid/ask orders. Because of the inquire effort of a large size of software-based order book can be significant and the great latency of data transfer between external memory and CPU, the latency of order book was decreased by three orders of magnitude compares to the software-based order book.

IV. CONCLUSION

In this paper, an FPGA-based high-frequency trading system is implemented on a Xilinx Kintex-115 1517-2e FPGA card. The system provides the functionality of network stack with TCP/IP, UDP/IP and ARP are supported. Financial protocol parsing based on the TAIFEX message protocol and tick-by-tick data protocol. A custom trading logic implemented on the system and is easy to integrate new algorithm with our infrastructure. A single product order book handling method is presented in this paper. The system specializes the application on Taiwan futures market to optimize the hardware efficiency and achieves an end-to-end approximate latency of 433 ns.

REFERENCES

- [1] J. W. Lockwood, A. Gupte, N. Mehta, M. Blott, T. English, and K. Vissers, "A low-latency library in FPGA hardware for high-frequency trading (HFT)," in *2012 IEEE20th Annual Symposium on High-Performance Interconnects*, vol. 1, 2012, pp. 9–16.
- [2] *Taiwan Futures Exchange TCP/IP TMP Messaging Specifications*. Taiwan Futures Exchange, 2021.
- [3] *Tick-by-tick Market Information Transmission Operation Manual*. Taiwan Futures Exchange, 2019.
- [4] Q. Tang, M. Su, L. Jiang, J. Yang, and X. Bai, "A scalable architecture for low-latency market-data processing on FPGA," in *IEEE*

Symposium on Computers and Communication (ISCC), vol. 1, 2016, pp. 597–603.

- [5] A. Boutros, B. Grady, M. Abbas, and P. Chow, “Build fast, trade fast: FPGA-based high-frequency trading using high-level synthesis,” in *2017 IEEE International Conference on ReConfigurable Computing and FPGAs (ReConFig)*, 2017, pp. 1–6.
- [6] A. Milanovic, S. Srbljic, and V. Struk, “Performance of UDP and TCP communication on personal computers,” in *2000 10th Mediterranean Electrotechnical Conference. Information Technology and Electrotechnology for the Mediterranean Countries. Proceedings. MeleCon 2000 (Cat. No.00CH37099)*, vol. 1, 2000, pp. 286–289 vol. 1
- [7] C. Leber, B. Geib, and H. Litz, “High frequency trading acceleration using FPGAs,” in *2011 21st International Conference on Field Programmable Logic and Applications*, 2011, pp. 317–322.